Multiplying & Dividing Fractions

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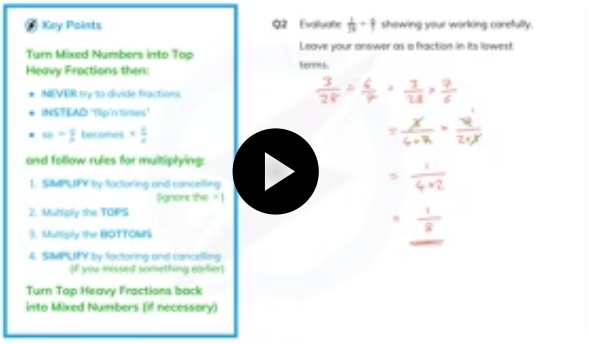

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Multiplying fractions

How do I multiply two fractions?

- **Cancel** any factors that are common to both the **numerator** and the **denominator**
  - The common factors could be in either of the fractions
  - $\frac{3}{25} \times \frac{10}{11} = \frac{3}{\cancel{25}^{5}} \times \frac{\cancel{10}^{2}}{11} = \frac{3}{5} \times \frac{2}{11}$
- **Multiply**
  - Multiply the **numerators**
  - Multiply the **denominators**
  - $\frac{3}{5} \times \frac{2}{11} = \frac{3 \times 2}{5 \times 11} = \frac{6}{55}$
- **Cancel** any common factors again if possible
  - There should be no common factors if you cancelled them all at the start

How do I multiply two fractions if one is a mixed number?

- Always convert **mixed numbers** into **improper fractions** before multiplying

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Worked Example

Find $\frac{4}{15} \times \frac{25}{11}$.

Answer:

The 15 and 25 can be cancelled before multiplying

$$\frac{4}{3 \times 5} \times \frac{5 \times 5}{11} = \frac{4}{3 \times \cancel{5}} \times \frac{\cancel{5} \times 5}{11} = \frac{4}{3} \times \frac{5}{11}$$

Multiply the numerators together and the denominators together

$$\frac{4 \times 5}{3 \times 11} = \frac{20}{33}$$

There is no further cancelling that can be done
Write down the fraction

$$\frac{20}{33}$$

Dividing fractions

How do I divide two fractions?

- **Flip** the second fraction and **change** ÷ into × "
  - $\frac{1}{3} \div \frac{4}{5}$ becomes $\frac{1}{3} \times \frac{5}{4}$
  - The 'flipped' fraction is called a **reciprocal fraction**
- **Cancel** any factors that are common to both the **numerator** and the **denominator**
- **Multiply** the fractions
  - Multiply the **numerators**
  - Multiply the **denominators**
- **Cancel** any common factors again if possible
  - There should be no common factors if you cancelled them all at the start

How do I divide two fractions when one of them is a mixed number?

- Always convert **mixed numbers** into **improper fractions** before dividing
  - **Convert** improper fractions **back** into mixed numbers at the end if required

Examiner Tips and Tricks

Worked Example

Divide $3\frac{1}{4}$ by $\frac{3}{8}$, giving your answer as a mixed number.

Answer:

Rewrite $3\frac{1}{4}$ as an improper fraction

$$3\frac{1}{4} = \frac{3 \times 4 + 1}{4} = \frac{13}{4}$$

Turn the division into a multiplication
'Flip' the second fraction and turn ÷ into ×

$$\frac{13}{4} \div \frac{3}{8} = \frac{13}{4} \times \frac{8}{3}$$

Cancel a factor of 4 from the numerators and denominators

$$\frac{13}{4} \times \frac{8}{3} = \frac{13}{\cancel{4}} \times \frac{\cancel{4} \times 2}{3} = \frac{13}{1} \times \frac{2}{3}$$

Multiply the fractions

$$\frac{13 \times 2}{1 \times 3} = \frac{26}{3}$$

Rewrite as a mixed number

$$\frac{26}{3} = \frac{24}{3} + \frac{2}{3} = 8\frac{2}{3}$$

$$8\frac{2}{3}$$

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